# Three Levers That Lower Validation Bits-per-Byte at a Fixed 300-Second Budget, and the Instrument Required to See Them

OPHIS Autonomous Research Campaign

*Single-node H200, frozen data pipeline, seed 42*

**Abstract**

We report six interventions measured on a fixed-*time* language-model training benchmark: a ∼50.3M-parameter GPT trained for 300 charged seconds on one H200 with the data pipeline frozen. Three lower validation bits-per-byte and three do not. The largest single effect is halving the short span of a sliding-window attention pattern; combined with a value-embedding density change and a second-moment reordering in the optimizer it produces the best model we have trained, improving val_bpb from 0.990345 to 0.986956. The combination realises most, but measurably not all, of the sum of its parts, and we locate the overlap in a shared throughput channel.

Our central methodological claim is that none of these effects is visible without a paired, device-counterbalanced design: the 70 byte-identical control runs in this campaign span a range of 0.034609, roughly an order of magnitude larger than the largest effect we report. We describe the design, the noise model motivating it, and—at length, because we believe it is the more transferable contribution—the instrument defects that had to be found and corrected before any number below could be trusted. Six of the seven defects repaired in this block were in our own apparatus rather than in the model.

## 1 Introduction

The benchmark fixes *time*, not tokens. A change that improves the model per step but costs throughput can be a net loss, and a change that buys steps by removing computation can be a net win even if it degrades the model per step. Throughput is therefore part of quality rather than a nuisance to be regressed away. We report raw val_bpb as the verdict everywhere and use step count only to explain *why* a result came out as it did.

The campaign began with no experimental priors; every inherited belief was deleted deliberately, on the grounds that most were wrong or unfalsifiable. What we had was a literature corpus and a set of properties of the apparatus. What we lacked—and had to build first—was a measuring instrument able to resolve the effects we sought.

**Contributions.** (i) Three confirmed levers and three confirmed non-levers, each measured under a counterbalanced design with activation independently verified (§4, §5). (ii) A combination experiment showing the levers are complementary but partly redundant, with the overlap channel identified (§6). (iii) An account of how the measurement apparatus failed, and what that implies for autonomous experimentation (§8).

## 2 Setup

A GPT of ∼50.3M parameters: vocabulary 8192, depth 8, $n_{\text{embd}}$ 512, MLP ratio 4, head dimension 128. Attention follows a repeating `SSSL` window pattern—three short-window layers and one long-window layer per block, final layer forced to full context. Value embeddings attach to every other layer; QK-norm is applied. The optimizer is Muon on 2D matrices with AdamW elsewhere. Training runs under `torch.compile` in bf16 for 300 charged seconds: on this host roughly 1000 steps at 524288 tokens per step, completing about two epochs of a fixed dataset.

The tokenizer, loader, packing, evaluator and BPB accounting are frozen and byte-identical to the upstream reference. Only the training file is edited. The host is shared; other tenants occupy most GPUs most of the time, and co-tenancy on a GPU invalidates a run.

## 3 The measurement problem

The 70 byte-identical control runs span 1.025517 at worst to 0.990908 at best—a range of 0.034609. Every effect we report is smaller than 0.004408. A single unpaired run therefore cannot see any of them.

The dominant nuisance is host CPU contention, which moves step count run-to-run through the frozen packing loop. Two structural facts follow.

**Concurrency, not proximity.** Runs must overlap in wall-clock time to share contention. Runs launched back-to-back on one GPU are sequential and the load drifts between them; grouping those as a "wave" reports a sequential spread under a within-wave label, flattering the instrument.

**The offset belongs to the GPU.** Our four devices have systematically different control means, with a fastest-to-slowest spread comparable to the effects being chased. An earlier version of this campaign attributed that offset to the CPU core block; correcting which factor owns it changed verdicts.

## 4 Design: the counterbalanced quad

Each experiment is two four-wide waves launched concurrently:

| | |
|---|---|
| wave A: | [ treat, ctrl, ctrl, treat ] |
| wave B: | [ ctrl, treat, treat, ctrl ] |

Across the pair the treatment occupies every device exactly once, as does the control. Deltas are formed *within* each device, removing that device's fixed offset inside each difference rather than relying on cancellation in the mean.

Two consequences matter. First, $n = 4$ pairings, so every $t$ carries 3 degrees of freedom and should be read as evidence of *direction*, not as a precise magnitude. Second, when the design breaks—a device set shifts between waves, or an arm crosses an operating-point boundary—the analysis **refuses a verdict** rather than averaging across a broken swap. This occurred three times in this block, costing three extra waves.

**Activation.** Every hypothesis declares a diagnostic the run must emit, checked *before* the outcome. A run whose diagnostic is absent, or which fails its rule, is **inconclusive** about the mechanism—never a negative. Otherwise a broken intervention and a genuine null are indistinguishable and good ideas are retired for free.

The instrument resolution, twice the pooled within-device control standard deviation divided by $\sqrt{2}$, is 0.000282 (pooled sd 0.000199). Effects below it are not claimed.

## 5 Results

All comparisons are same-device paired within a single counterbalanced quad pair, seed 42.

| intervention | mean Δ | verdict |
|---|---|---|
| `swdiv` 2→4 | −0.002298 | better |
| `ve` 2→1 | −0.001487 | better |
| `precond` | −0.001147 | better |
| `qk_suppress` 0.1 | +0.001226 | *inconclusive* |
| `unet` | +0.004123 | worse |
| `tbs` 19→20 | +0.022545 | worse |

### 5.1 Halving the short attention span

At sequence length 2048 the `SSSL` pattern gives six short-window and two long-window layers. Halving the short window from 1024 to 512 takes per-token FLOPs from 239.078 to 220.204: a removal of 18.874, or 7.9% of the total—*not* the 26% that attention as a whole represents. The two long-window layers hold about 40% of the attention term and are untouched.

This distinction was not academic. The hypothesis was originally justified with the 26% figure, overstating the throughput lever by ∼3.3×. The literature review caught the error before the experiment ran; the arithmetic was corrected and the prediction threshold was *deliberately not lowered* to compensate. The corrected, harder prediction was met anyway.

Step counts rose in every cell and quality held. This is the opposite outcome to an earlier arm reducing Newton–Schulz iterations, worth ∼0.2% of step time, which bought no steps and came back worse: 7.9% is enough to move the critical path and 0.2% is not.

**A correction we owe the reader.** We first reported this effect as −0.002463 at $t = -20.3$. An independent audit found that estimate stitched pairs from two experiments run 1.4 hours apart. The clean single-experiment quad gives −0.002298 (sd 0.000142, $t = -32.4$). The stitched figure overstated the effect by about 7%; the corrected value is used throughout.

### 5.2 Value embeddings on every layer

`ve` is a *stride*, so `ve=1` is denser: tables go from four to eight and parameters from 50.3M to 67.1M. This 33% larger model ran slower and completed *fewer* steps than its control in every cell, and won regardless—quality gained per step more than repays throughput lost.

This contradicts the prior we registered before running it. Three corpus claims opposed it: all-layer injection measured worse than single-layer, dense per-layer expansion saturating, and evidence that such memories' benefit is the gated pathway rather than the learned table content. All three carried honestly low transfer scores (image autoregression; tokenizer-transfer fine-tuning; a 2.5B mixture-of-experts). They retain their opposing stance; what we now know is that they do not carry to 50M parameters at 300 seconds.

**One void cell.** On the slowest device the added parameters cost enough throughput to finish at 501.2M tokens and epoch 1.0 against its control's 2.0—a different operating point, so the pair is excluded. The exclusion rule is structurally biased against parameter-adding treatments; here it cut *against* the result, since including the cell would have strengthened it. This leaves `ve` with three pairings and 2 degrees of freedom, the weakest of our three wins.

### 5.3 Second-moment reordering

Moving the NorMuon second-moment computation before the polar iteration, with unit-RMS momentum normalisation, gives −0.001147 across four devices. The activation observable separates the arms by roughly two orders of magnitude, because the second moment is taken over raw momentum rather than an orthogonalized update.

### 5.4 Two clean negatives

**Zero-initialised U-net skips** raised val_bpb by 0.004123 with a four-device standard deviation of only 0.000078—the tightest measurement in the campaign. That precision follows from the intervention: the skips add one elementwise addition per lower-half layer and no matrix multiply, so they cannot buy or cost steps, and the step counts confirm they did not. The effect is pure quality, free of the throughput entanglement that explains most variance elsewhere. The gate parameters grew from exactly zero to $\sim$0.2, so the model *used* the path and was worse for it. This intervention was the independent top pick of two separate proposers in the same council round; their agreement carried no evidential weight whatever.

**Doubling tokens per step** raised val_bpb by 0.022545, roughly eighty times the resolution and by far the largest effect measured. Step counts halved as expected. At fixed time this model sits far *above* its critical batch size, so halving the optimizer update count costs far more than reduced gradient noise returns. The step law cannot explain this away: it was fitted at 524288 tokens per step and this arm moves that very quantity.

### 5.5 One honest inconclusive

`qk_suppress` produced a textbook-looking counterbalanced negative, $+0.001226$ at $t = +15.8$. We do *not* report it as a negative. Its declared diagnostic reads exactly 1.0 in every treatment *and* every control, because QK-norm pins query RMS to unit norm by construction. It could never have shown engagement, so the run cannot distinguish "the mechanism works badly" from "the edit never bound and something else moved". The hypothesis's own pre-registered falsifier named this case and we apply it against ourselves. The axis is untested, not refuted.

## 6 Combining the levers

The three winners were run together against the *unchanged* control:

$$\Delta = -0.004408, \quad \mathrm{sd} = 0.000152, \quad \mathrm{sem} = 0.000076$$

over four devices, all eight arms at epoch 2.0, with all three components independently confirmed active in the same records. The best model is 0.986956.

The naive sum of the three separately measured effects is $-0.004932$. The combination realises about 89% of it, a shortfall of 0.000524.

**Where the overlap is.** `ve` and `swdiv` both act partly through step count, in opposite directions: `ve` costs throughput by adding parameters, `swdiv` buys it by removing FLOPs. Stacked, they nearly cancel in that channel—on the slowest device the combined arm ran 991 steps against its control's 992, where `ve` alone managed 956 and voided. Some of the two levers' separate gains were the same gain counted twice.

**Why this reading is licensed.** The interaction was predicted *before* the run and the reading order fixed in advance: check epoch on every arm first, then look at val_bpb. Had the combined arm voided on the slow device, a shortfall would have been an operating-point artefact saying nothing about redundancy. It did not void, and the step count shows why. After the fact the two explanations are indistinguishable, which is the entire reason the order was pre-committed.

**On the statistics.** Our reported $t = -58$ rides a 3-df variance estimate about half the control-derived expectation; a chi-square check gives $p \approx 0.14$, so this is not evidence of genuinely lower treatment noise, and a fairer $t$ is $\approx -29$. The win is unambiguous either way. We initially justified the shortfall as "2.4$\times$ resolution", which repeats a borrowed-constant error our own lesson log condemns: the additive sum is itself three noisy estimates and its error must be propagated. Doing so properly gives $\approx$3.3 standard errors at Welch df $\approx 8.9$, so sub-additive redundancy stands—but by correct arithmetic, not the argument we first gave.

**Two caveats an audit raised after first submission.** First, `precond`'s $t = -12.3$ is a same-device paired statistic computed *across* waves; the raw within-wave paired $t$ is $\approx -2.0$. Both share the mean $-0.001147$, and the same-device pairing is defensible because it removes the device term inside each difference rather than leaving it to cancel—but it relies on the device model being stable, so it is model-adjusted rather than raw, and we state it as such. Second, the configuration string `precond="pre"` names *two* different implementations in our records, the pre-repair version whose second-moment buffer collapsed onto its clamp floor and the repaired one; only the latter supports the number above, and our analysis keys arms by generated-source hash precisely so the two are never pooled.

## 7 Threats to validity

**Single seed.** Every result is seed 42 on one host, by operator direction to conserve wall clock. Nothing here is established as seed-general.

**Degrees of freedom.** All $t$ statistics carry 3 df or fewer.

**Device advantage.** The headline best model ran on our fastest device, which carries a $\sim$0.0025 advantage over the slowest. The *effects* are device-counterbalanced and unaffected; the headline *number* is flattered.

**Instrument defects dominate.** Of 47 registered lessons, 8 are scientific negatives and the remainder are process or measurement failures.

## 8 What the instrument cost

We describe these because they are more transferable than the results, and because each was found by audit rather than intuition.

**An instrument that could not measure what it claimed.** A GPU-slack metric built from CUDA events bracketing the step. Such a pair measures elapsed time between two stream markers *including* idle intervals, and our bracket enclosed the data loader, pinning the quantity to step time by construction.

It read near-zero slack on every run, which would have been reported as "the GPU is saturated".

**A metric silently discarded from every run.** The per-token FLOP estimate was printed by every variant and dropped by a parser requiring whitespace after a colon where the print emits none—absent from all 116 records. Recovered by re-reading each run's own log and verified to reproduce from first principles.

**Three vacuous activation diagnostics.** In one session we shipped three hypotheses whose diagnostic the control also satisfied, all registered *after* the lesson recording that failure mode existed. Prose did not prevent recurrence; a pre-registration check testing a proposed diagnostic against the existing control corpus does.

**A policy that strangled the search.** The explore/exploit controller retired an axis after four consecutive non-improving runs. A counterbalanced quad is four runs of the *same* value—so the moment we adopted the design that makes results trustworthy, every axis began closing after one experiment. Three axes closed having tried exactly one distinct value, including one that closed on evidence its *opposite* direction was worth testing.

**A ledger that could not reach its own threshold, then looked forward in time.** The explore/exploit share was first measured over runs; since each experiment is one novel run plus $\sim$7 counterbalancing replicates, the exploitation share could not exceed $\sim$12.5% against a 30% floor. Corrected to per-experiment, it then judged whether an axis "pays" using *final-state* knowledge, retroactively classifying the runs that *discovered* each lever as exploitation of it, and reporting a balanced 50% where the true chronological share was 1/8. Three defects, in a ledger built specifically to keep us honest about that number; each time the measurement flattered the thing it measured.

**A gate that could never be satisfied.** Adoption of the confirmed stack was deferred pending a second-seed replication—a condition the operator's own single-seed directive precludes. An audit named this a pocket veto and priced it: sixteen queued runs measured against a control 0.004408 worse than the demonstrated configuration. A gate that cannot be satisfied is not caution; it is a refusal wearing caution's clothing, and it is worse than an explicit refusal because it looks like a pending decision and is never revisited.

**The pattern.** Every one of these fixes was discovered when a rule blocked something we wanted to run. We have never once gone looking for a rule that was too *permissive*. That asymmetry is itself a threat to validity, and we do not know how to correct it from the inside—which is the strongest argument we can make for independent, adversarial audit as a standing component of autonomous research rather than an occasional one.

## 9 Conclusion

Three interventions lower val_bpb at a fixed 300-second budget and combine to about 89% of their additive sum, improving the best model from 0.990345 to 0.986956. Three others do not, one by a very large margin. The measurement design making these statements possible—concurrent yoked pairs, device counterbalancing, pre-registered activation diagnostics, and a willingness to refuse a verdict when the design breaks—mattered more than any individual intervention, because the effects are roughly a twentieth of the noise an unpaired comparison would face.